COSC2673 | COSC2793 (Computational) Machine Learning

Assignment 1

Name: Vadhthanak Vibol ID: s3951117

Due date and time: Wednesday, 15 April 2026, 11:59 pm AET (Week 6).

```
# Importing section

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression, Ridge, Lasso
from sklearn.metrics import mean_squared_error, mean_absolute_error,
r2_score
from sklearn.preprocessing import StandardScaler
```

Data Preparation

Load in Data for pandas to read the dataset

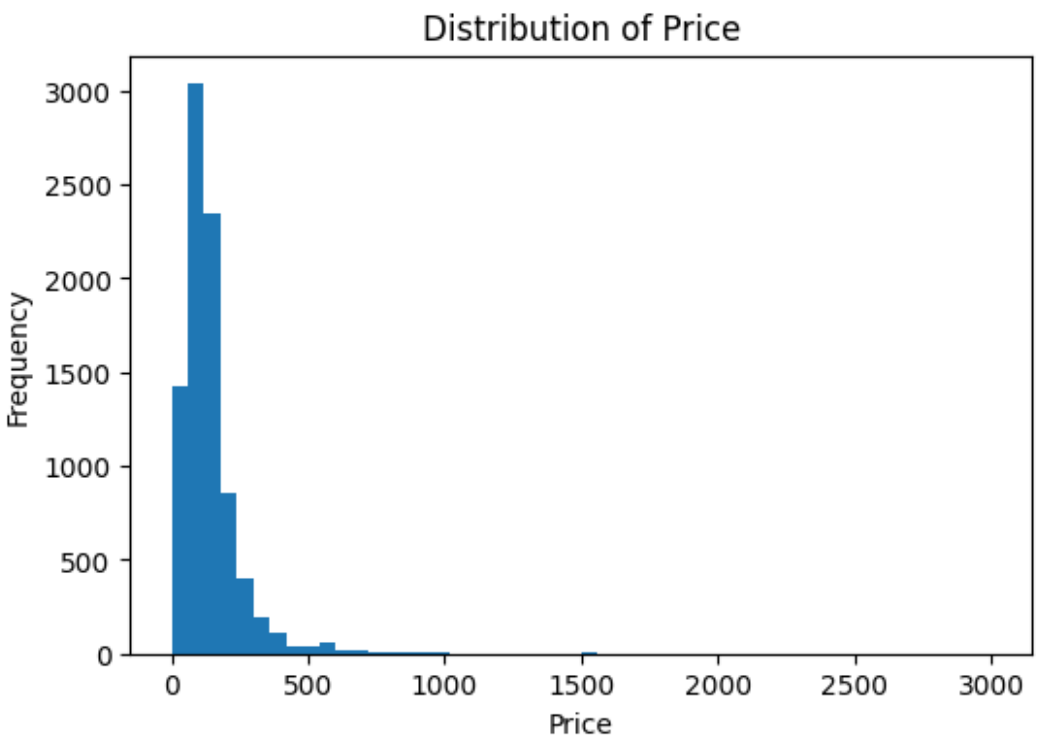
```
# Load data

train = pd.read_csv('./dataset/train_data.csv')
test = pd.read_csv('./dataset/test_data.csv')
```

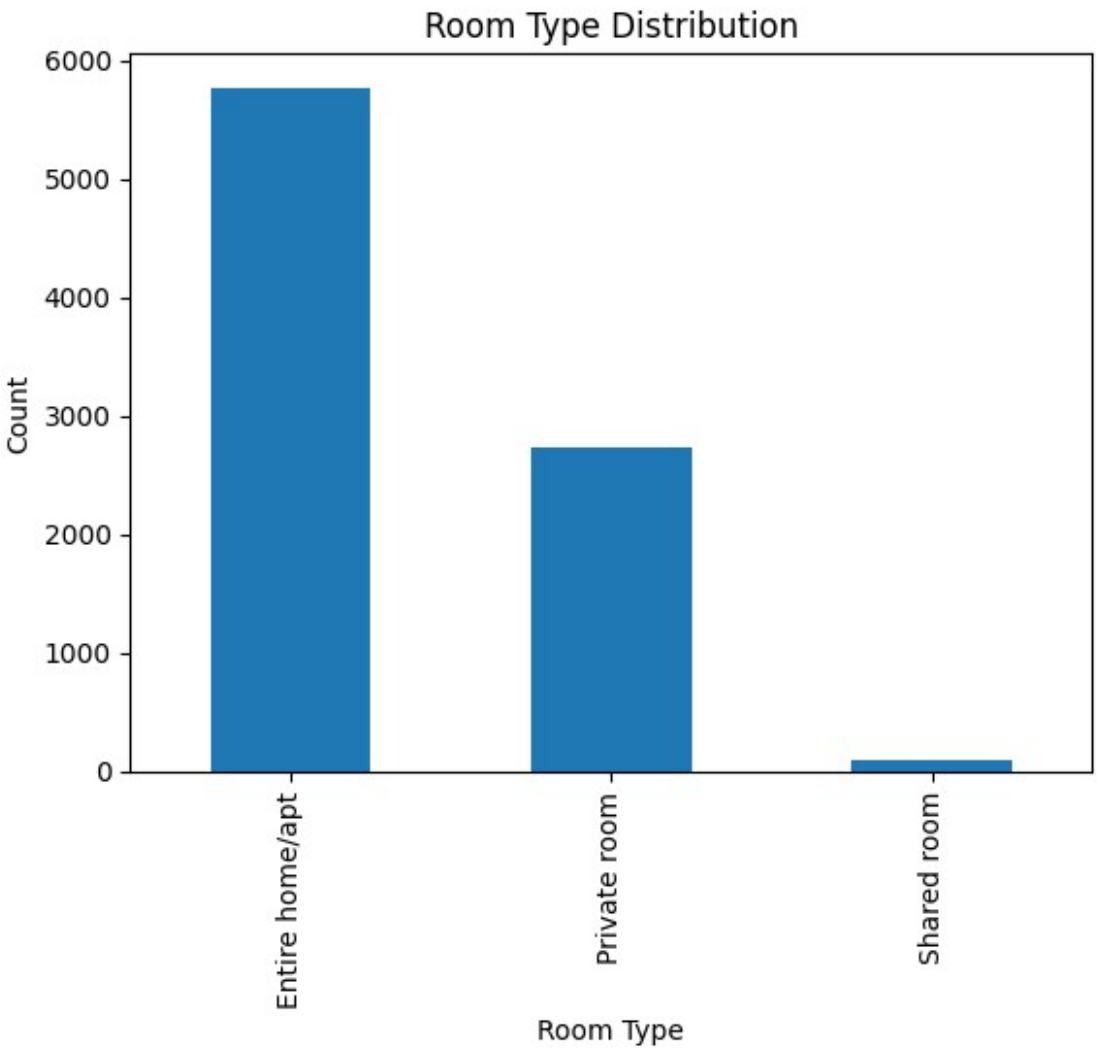
Exploratory Data Analysis

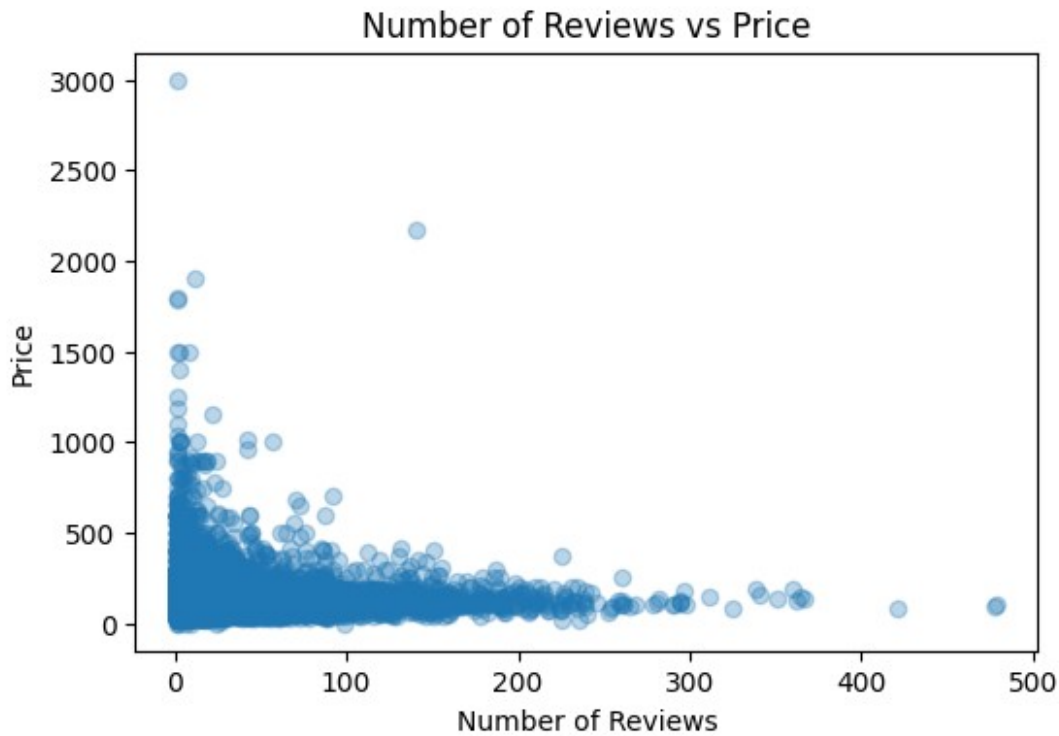
The brief data analysis will happen here where we will briefly compare the frequency of the AirBnB properties

```
# Distribution of price
plt.figure(figsize=(6,4))
plt.hist(train["price"], bins=50)
plt.title("Distribution of Price")
plt.xlabel("Price")
plt.ylabel("Frequency")
plt.show()
```

```
# Room type distribution
train["room_type"].value_counts().plot(kind='bar')
plt.title("Room Type Distribution")
plt.xlabel("Room Type")
plt.ylabel("Count")
plt.show()


# Reviews vs price
plt.figure(figsize=(6,4))
plt.scatter(train["number_of_reviews"], train["price"], alpha=0.3)
plt.title("Number of Reviews vs Price")
plt.xlabel("Number of Reviews")
plt.ylabel("Price")
plt.show()
```




Room Type Distribution
6000
5000
4000
3000
2000
1000
0
Count
Entire home/apt
Private room
Shared room
Room Type



The distribution of price is highly skewed, with a few very high values. This suggests the presence of outliers, which may negatively affect model performance.

The price distribution is highly skewed to the lower end of the graph. This supports the use of log transformation to improve model performance.

```
# Drop the country column
train = train.drop('country', axis=1)
test = test.drop('country', axis=1)

# Convert boolean columns
for col in ["host_is_superhost", "instant_bookable"]:
    train[col] = train[col].map({"t": 1, "f": 0})
    test[col] = test[col].map({"t": 1, "f": 0})
```

Removing unecessary columns and convert true/false columns to 1s and 0s to make data readable

```
y = train["price"]
X = train.drop(columns=["price"])
```

Model Performance Before Preprocessing

```
# ========================
# BEFORE PREPROCESSING
# ========================
```

```python
# Split X and y
y_raw = train["price"]
X_raw = train.drop(columns=["price"])

# One-hot encoding
X_raw = pd.get_dummies(X_raw, columns=["city", "room_type"])

# Train-validation split
X_train_b, X_val_b, y_train_b, y_val_b = train_test_split(
    X_raw, y_raw, test_size=0.2, random_state=42
)

# Models
models_before = {
    "Linear": LinearRegression(),
    "Ridge": Ridge(),
    "Lasso": Lasso(alpha=0.1)
}

results_before = {}

for name, model in models_before.items():
    model.fit(X_train_b, y_train_b)
    y_pred = model.predict(X_val_b)

    results_before[name] = {
        "RMSE": np.sqrt(mean_squared_error(y_val_b, y_pred)),
        "MAE": mean_absolute_error(y_val_b, y_pred),
        "R2": r2_score(y_val_b, y_pred)
    }

results_df_before = pd.DataFrame(results_before).T
results_df_before
```

```
              RMSE        MAE        R2
Linear  103.683616  47.816145  0.395486
Ridge   103.669416  47.713224  0.395651
Lasso   103.614379  47.220241  0.396293
```

R2 Analysis

Before preprocessing the model data is messy and scattered making the Coefficient of Determination low and the graph will be underfitting. If the model were to be chosen it would be a Lasso Regression training model since it has the highest Coefficient of Determination amongst others

Model Performance After Preprocessing

```
# ========================
# AFTER PREPROCESSING
# ========================

# Remove outliers
train_clean = train[train["price"] < train["price"].quantile(0.99)]

# Split X and y
y = train_clean["price"]
X = train_clean.drop(columns=["price"])

# Log transform
y = np.log1p(y)

# One-hot encoding
X = pd.get_dummies(X, columns=["city", "room_type"])

# Train-validation split
X_train, X_val, y_train, y_val = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Scale after split
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_val = scaler.transform(X_val)

# Models
models_after = {
    "Linear": LinearRegression(),
    "Ridge": Ridge(),
    "Lasso": Lasso(alpha=0.1)
}

results_after = {}

for name, model in models_after.items():
    model.fit(X_train, y_train)
    y_pred = model.predict(X_val)

    results_after[name] = {
        "RMSE": np.sqrt(mean_squared_error(y_val, y_pred)),
        "MAE": mean_absolute_error(y_val, y_pred),
        "R2": r2_score(y_val, y_pred)
    }

results_df_after = pd.DataFrame(results_after).T
results_df_after
```

```
            RMSE       MAE        R2
Linear  0.350451  0.261792  0.647628
Ridge   0.350449  0.261791  0.647634
Lasso   0.394810  0.300445  0.552779
```

R2 Analysis

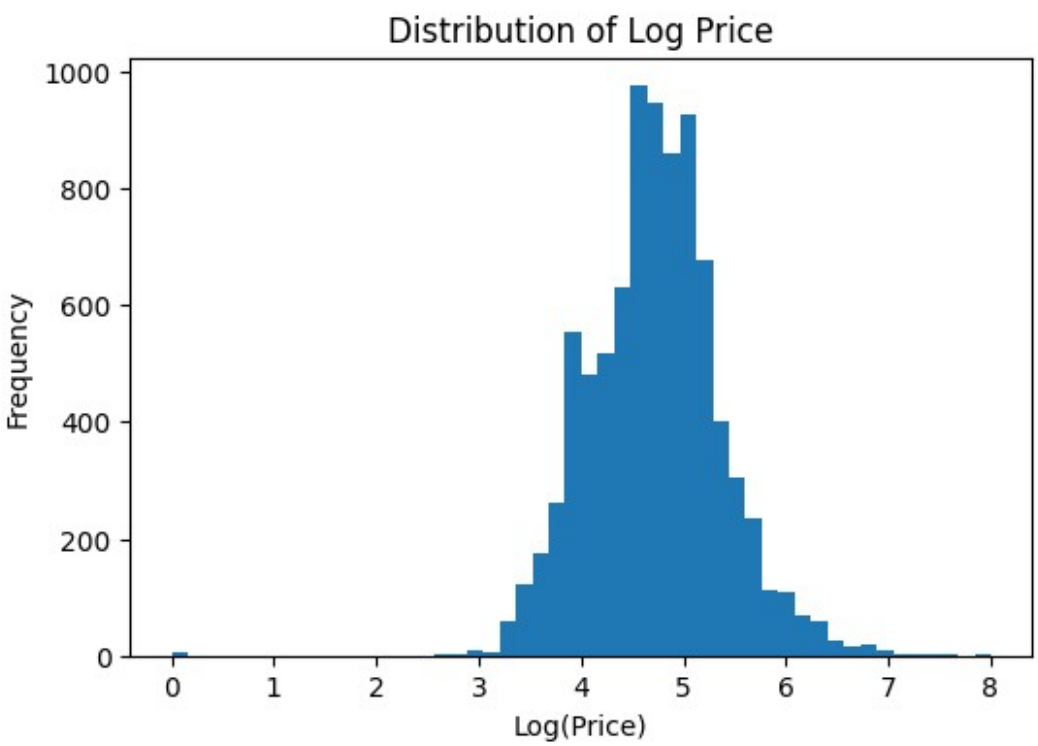
After thoroughly preprocessing data, it has become much more clean and structured helping the model train better and the data to be modeled to be much better structured to be used.

In a turn of events, the R2 numbers are two times higher making it a better fitting model to train, where the Ridge Regression model beats Linear Regression model by a hair by having a Coefficient of Determination higher than Linear.

Being the better model, it will be used as the training model to be more accurate with predcitions.

Data Distribution After Preprocessing

```
plt.figure(figsize=(6,4))
plt.hist(np.log1p(train["price"]), bins=50)
plt.title("Distribution of Log Price")
plt.xlabel("Log(Price)")
plt.ylabel("Frequency")
plt.show()
```



Comparison of Results

Comparision of the two before and after preprocessing grouped up into one single box

Ridge regression performed best due to better generalisation from regularisation.

```
before_df = results_df_before.copy()
after_df = results_df_after.copy()

before_df["Stage"] = "Before"
after_df["Stage"] = "After"

comparison = pd.concat([before_df, after_df])
comparison
```

```
              RMSE        MAE        R2   Stage
Linear  103.683616  47.816145  0.395486  Before
Ridge   103.669416  47.713224  0.395651  Before
Lasso   103.614379  47.220241  0.396293  Before
Linear    0.350451   0.261792  0.647628   After
Ridge     0.350449   0.261791  0.647634   After
Lasso     0.394810   0.300445  0.552779   After
```

Final Selection

Comparing the two results we can see that after a much needed preprocessing on the data as raw data is often noisy, inconsistent, incomplete, or incorrectly formatted for machine learning models. The data is more consistent with trends and is easier to model with a better fitting line.

Ridge regression is chosen for the training model as it has the best fit as R2 is the highest compared to others after preprocessing.

```
# Ridge Regression Model Training

# Clean data again
train_clean = train[train["price"] < train["price"].quantile(0.99)]

# Create a new X and y for training
y_final = np.log1p(train_clean["price"])
X_final = train_clean.drop(columns=["price"])

# One hot encoding
X_final = pd.get_dummies(X_final, columns=["city", "room_type"])
test_processed = pd.get_dummies(test, columns=["city", "room_type"])

# Match columns
X_final, test_processed = X_final.align(test_processed, join='left',
axis=1, fill_value=0)

# Scale
```

```
scaler = StandardScaler()
X_final_scaled = scaler.fit_transform(X_final)
test_scaled = scaler.transform(test_processed)

# Train final Ridge model
final_model = Ridge()
final_model.fit(X_final_scaled, y_final)
```

```
Ridge()
```

```
# Predict test data
test_pred = final_model.predict(test_scaled)

# Convert back from log scale
test_pred = np.expm1(test_pred)

test_pred[:10]
```

```
array([ 48.19522802, 127.37691938,  63.45233416,  46.55427327,
        69.25519401,  59.86444025, 156.85232499, 132.29559491,
        62.88941821, 135.56553731])
```

```
submission = pd.DataFrame({
    "price": test_pred
})

submission.to_csv("S3951117_submission.csv", index=False)

submission.head()
```

```
        price
0   48.195228
1  127.376919
2   63.452334
3   46.554273
4   69.255194
```

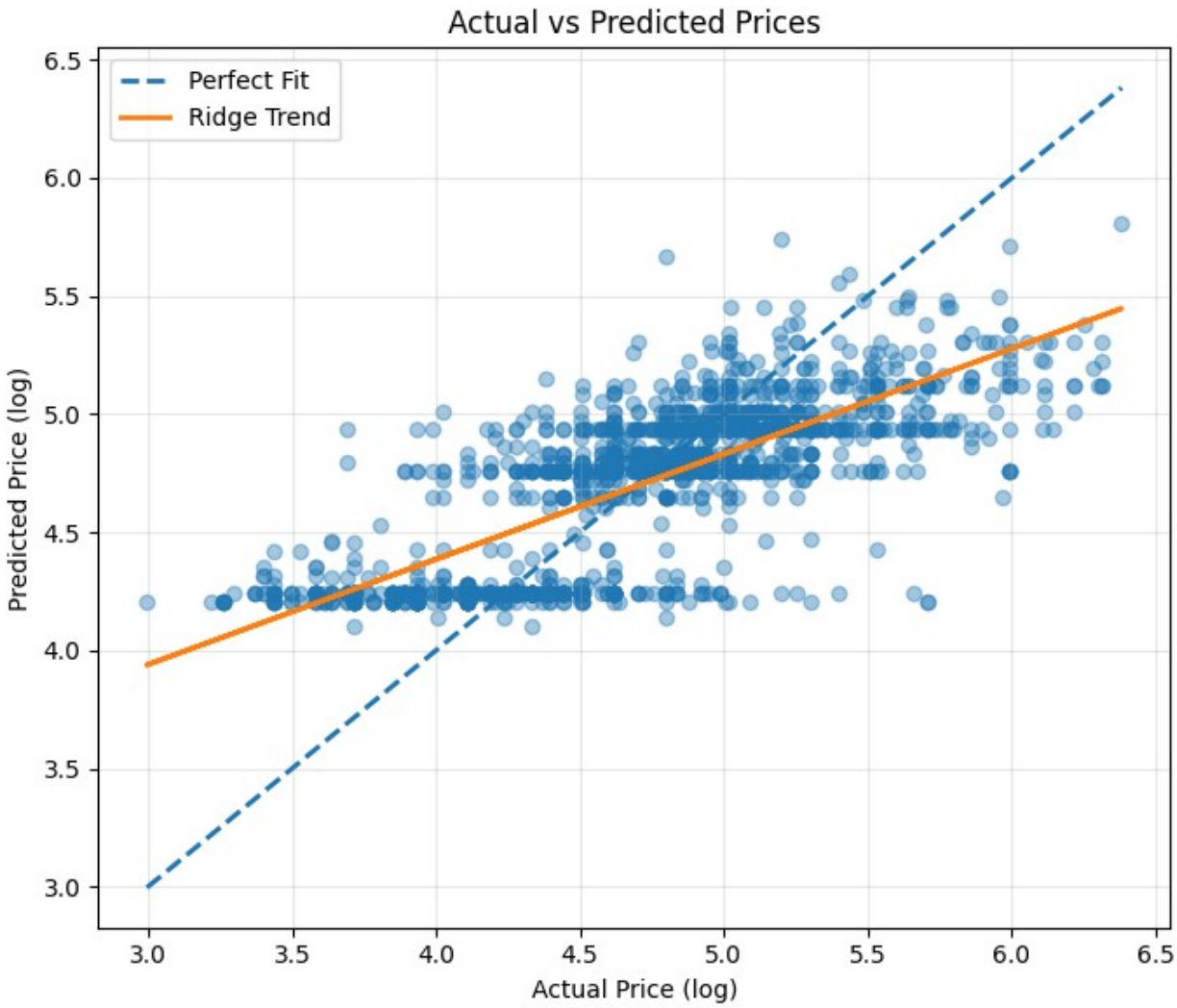
Final Model Training - Ridge

```
# Cleaning Data
train_clean = train[train["price"] < train["price"].quantile(0.99)]

y = np.log1p(train_clean["price"])
X = train_clean.drop(columns=["price"])

X = pd.get_dummies(X, columns=["city", "room_type"])
test_processed = pd.get_dummies(test, columns=["city", "room_type"])

# Make columns match
X, test_processed = X.align(test_processed, join='left', axis=1,
fill_value=0)
```

```python
# Scale
scaler = StandardScaler()
X = scaler.fit_transform(X)
test_scaled = scaler.transform(test_processed)

# Train final model
final_model = Ridge()
final_model.fit(X, y)
```

```
Ridge()
```

```python
import numpy as np
import matplotlib.pyplot as plt

plt.figure(figsize=(7,6))

# Scatter
plt.scatter(y_val, y_pred, alpha=0.4)

# Perfect line (y = x)
plt.plot(
    [y_val.min(), y_val.max()],
    [y_val.min(), y_val.max()],
    linestyle='--',
    linewidth=2,
    label='Perfect Fit'
)

# Ridge trend line
m, b = np.polyfit(y_val, y_pred, 1)
plt.plot(
    y_val,
    m*y_val + b,
    linewidth=2,
    label='Ridge Trend'
)

plt.xlabel("Actual Price (log)")
plt.ylabel("Predicted Price (log)")
plt.title("Actual vs Predicted Prices")

plt.legend()
plt.grid(alpha=0.3)

plt.tight_layout()
plt.show()
```



The model may not capture all real-world factors such as property condition or seasonal demand.